25/07/26 Co-2 Assessment - 2

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(1) Suppose that the data for analysis includes the attribution age. The age value of data tuples are (in increasing order)

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33

35, 35, 35, 35, 36, 40, 45, 46, 52, 70

(a) What is the mean of the data? what is the median?

$$\text{Mean} = \frac{1}{N}\sum_{i=1}^{n} x_i$$

$$= \frac{809}{27} = 29.96$$

= App 30.00

$$\text{Median} = \frac{n+1}{2} = \frac{28}{2} = 14 \rightarrow \text{poistion}$$

Value = 25

(b) What is the mode of the data? Comment on the data modify (i.e, bimodal, trimodal etc)

25, 35 are repeating 4 times so they are bimodal

(c) what is the midrange of the data?

$$\text{Midrange} = \frac{\text{max}+\text{min}}{2} = \frac{70+13}{2} = 41.5$$

(d) Can you find (roughly) the First quartile (Q_1) and third quartile (Q_3) of the data?

$q_1 = 20$ $q_3 = 35$

Sol Here 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25

25

$q_1 = 20$

30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

$q_3 = 35$

$= q_3 - q_1$

$= 35 - 20$

$= 15$

(e) Give the Five Number Summary of the data

Sol Min, q_1, median, q_3, max

= min = 13

$q_1 = 20$

median = 25

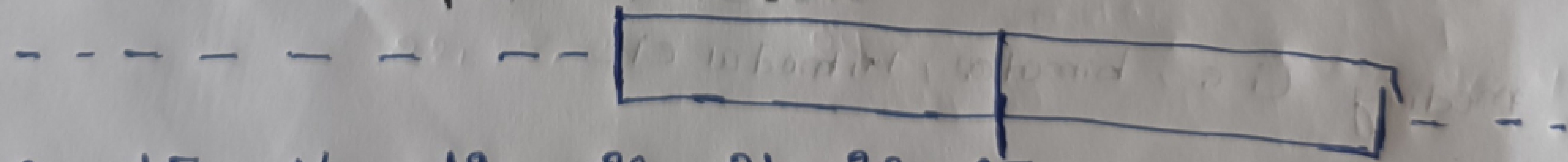
$q_3 = 35$

max = 70

(13, 20, 25, 35, 70)

(f) Show a boxplot of the data

Sol



13, 15, 16, 19, 20, 21, 22, 25, 30, 33, 35, 36

--- 40, 46, 52, 70

(2) Given the following measurment for the variable age:
18, 22, 25, 42, 28, 43, 33, 35, 56, 28 standardize the variable by the following :-

Compute the mean absolute deviation of age

Sol Mean $\frac{1}{N} \sum_{i=1}^{n} x_i$ $n=10$

$$= \frac{18+22+25+42+28+43+33+56+28}{10}$$

$= 33$

Find mean absolute deviation of age

Age	$\lvert x - \bar{x} \rvert$
18	$18-33 = 15$
22	$22-33 = 11$
25	$25-33 = 8$
42	$42-33 = 9$
28	$28-33 = 5$
43	$43-33 = 10$
33	$33-33 = 0$
35	$35-33 = 2$
56	$56-33 = 23$
28	$28-33 = 5$
	88

$= \frac{88}{10}$

$= 8.8$

(3) Suppose that the data for analysis includes the attributes age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. use smoothing by bin means to smooth the above data using a bin depth of 3. Illustrate your step.

Sol

n = 27

No of bins 8/7/8

bin depth = 3

$$\text{Total No of bins} = \frac{\text{Total No of values}}{\text{bin depth}}$$

$$= \frac{27}{3}$$

$$= 9$$

Bin	origin value	mean	Smooted value
B1	13, 15, 16	14.67	14.67, 14.67, 14.67
B2	16, 19, 20	59	59, 59, 59
B3	20, 21, 22	21	21, 21, 21
B4	22, 25, 25	24	24, 24, 24
B5	25, 25, 30	40	40, 40, 40
B6	33, 33, 35	33.6	33.6, 33.6, 33.6
B7	35, 35, 35	35	35, 35, 35
B8	36, 40, 45	40.33	40.33, 40.33, 40.33
B9	46, 52, 70	56	56, 56, 56

(4) use the two method below to normalize the following group of data 200, 300, 400, 600, 1000 (a) min-max normalization by setting min = 0 and max = 1 (b) z-score normalization

Sol

Min-Max Normalization

$$\tilde{x} = \frac{x - min}{Max - min}$$, Min = 200, max = 1000

Value	$\tilde{x}$	
200	(200 - 200) / 800	= 0
300	(300 - 200) / 800	= 0.125
400	(400 - 200) / 800	= 0.250
600	(600 - 200) / 800	= 0.500
1000	(1000 - 200) / 800	= 1.000

$\boxed{0, 0.125, 0.250, 0.500, 1.00}$

Z-Score Normalization

$$z = \frac{x - \mu}{\sigma}$$

$$\mu = \frac{200 + 300 + 400 + 600 + 1000}{5}$$

$$= \frac{2500}{5} = 500$$

Standard deviation

x	$x - \mu$	$(x-\mu)^2$
200	-300	90,000
300	-200	40,000
400	-100	10,000
600	100	10,000
1000	500	250,000
		4,00,000

$$\sigma^2 = \frac{400000}{5} = 80000$$

$$\sigma = \sqrt{80000} = 282.84$$